

# Assignment 6: OpenClaw Agent for Text Collection and Analysis (15 Points; DDL April 30th)

CSS5120

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## 1 Overview

In this assignment, you will revisit the **same task** that you completed in the earlier LLM-agent assignment, but this time you will reimplement it using **OpenClaw**. The goal is not to design a new NLP task. Instead, the goal is to keep the topic, task objective, and overall workflow target as consistent as possible, so that you can make a meaningful comparison between two different ways of building an agent-based workflow.

This assignment therefore has two connected purposes. First, it gives you practical experience using OpenClaw as a platform for building an NLP-related workflow. Second, it asks you to compare this OpenClaw-based implementation with your earlier self-built LLM-agent workflow. By keeping the task as similar as possible, you will be able to reflect more clearly on the differences in design, execution, flexibility, usability, and output quality.

As before, the workflow should form a small but complete NLP pipeline. Your OpenClaw workflow should involve text collection or preparation, at least one meaningful NLP analysis stage, and a final output stage. In addition, your report should include a comparison between the OpenClaw version and your earlier LLM-agent version, focusing on both the **workflow process** and the **final output**.

## 2 Task Requirements

### 2.1 Reuse the Same Task from the Earlier Assignment

For this assignment, you should reuse the **same topic and task setting** from your earlier LLM-agent assignment as much as possible. This includes the same general task objective, a similar data source if feasible, and the same or highly comparable NLP analysis goal. The purpose is to make the comparison meaningful. If the topic, data, and analysis target all change substantially, then the final comparison will be much less informative.

At the same time, this does **not** mean that you may simply resubmit your previous work. The workflow must be genuinely reimplemented through OpenClaw. You should use OpenClaw as the main system through which the task is organized and executed.

### 2.2 Reimplement the Workflow in OpenClaw

You should build an OpenClaw-based workflow that addresses the same task as your previous assignment. The workflow should still include data collection or preparation, at least one NLP analysis stage, and a final output stage. However, these stages should now be organized through OpenClaw rather than through your previous self-built notebook or script-based workflow.

In other words, OpenClaw should function as the main system through which the workflow is structured and executed. It is not enough to use OpenClaw only as a simple chat interface while keeping the actual workflow almost entirely outside the platform.

You may use OpenClaw features such as skills, scheduled tasks, channels, or other available capabilities if they are useful for your design. The exact implementation is flexible, but the final system should clearly function as an OpenClaw-based workflow rather than as a simple one-shot prompt experiment.

If your workflow uses one or more **skills**, you should be able to explain what role they play in the workflow. For example, you should consider whether a skill is used for retrieval, organization, parsing,

summarization, or another part of the task, and whether the agent is actually able to invoke it correctly in practice.

## 2.3 Collect and Prepare Textual Data

As in the earlier assignment, the workflow should involve text-based data and should contain at least 100 text items. The texts should be short enough to process conveniently in your environment. You should clearly describe where the texts came from and how OpenClaw was involved in collecting, filtering, organizing, or preparing them.

Because the goal of this assignment is comparison, you should try to keep the data source and collection logic as similar as possible to your earlier assignment. Minor differences are acceptable if they arise naturally from timing, platform behavior, or tool availability, but the overall task should remain comparable.

## 2.4 Complete at Least One NLP Analysis Task

After the texts have been collected, your OpenClaw workflow should perform at least one meaningful NLP task. Suitable tasks include sentiment analysis, topic or theme discovery, stance analysis, keyword extraction, summarization, or another reasonable text analysis problem. You may complete more than one task if your workflow remains manageable, but one clear and well-executed task is sufficient.

The purpose here is not to maximize complexity, but to allow a fair comparison with your earlier assignment. You should therefore focus on making the OpenClaw version coherent, interpretable, and comparable to your previous implementation.

## 2.5 Show the Workflow and Output

Your submission should clearly demonstrate how the OpenClaw workflow operates. Another reader should be able to understand what data were used, how OpenClaw handled the workflow, what analysis was performed, and what result was produced.

At least one part of the submission should show the workflow in action. This may take the form of a screenshot, a task configuration, a prompt used inside OpenClaw, an execution trace, a scheduled task setup, or another concrete example that makes the workflow visible and understandable.

## 2.6 Compare the OpenClaw Version with the Earlier LLM-Agent Version

A central part of this assignment is the comparison between the two implementations. You should discuss both the **process** and the **output**. In terms of process, you may reflect on issues such as workflow design, ease of setup, flexibility, controllability, observability, reliability, or support for recurring tasks. In terms of output, you may compare the quality, structure, usefulness, or consistency of the final results.

The purpose of this comparison is not to declare that one approach is always better. Instead, you should try to understand what changes when the same task is implemented through OpenClaw rather than through a self-built LLM-agent workflow.

## 2.7 Reflection

You should also briefly reflect on what you learned from this comparison. For example, you may discuss whether OpenClaw made the workflow easier to structure, whether it introduced new constraints, whether it helped with automation, and what kinds of tasks seem better suited to OpenClaw or to self-built workflows.

# 3 A Small Example

Suppose that in your earlier assignment you built a workflow for collecting public comments about an AI product, organizing them, and then performing sentiment analysis and theme discovery. In this assignment, you would keep that same task and reimplement it in OpenClaw. You would then compare the OpenClaw version with the earlier version in terms of how the workflow is organized, how easy it is to run, how well the system handles repeated execution, and whether the final analysis output differs in useful ways.

This example is only intended to illustrate the assignment idea. You should use your own earlier task rather than copying this example literally.

## 4 Report Requirements

Your report should clearly explain what task you reused from the earlier assignment, how you reimplemented it in OpenClaw, what text data were used, what NLP task or tasks were performed, and what results were obtained. The report should also include a dedicated comparison with your previous LLM-agent workflow.

At minimum, the report should allow the reader to understand the reused task setting, the OpenClaw workflow structure, the analysis task, the final output, and your reflections on the differences between the two implementations. When appropriate, you may include a figure, a table, a screenshot, or a short execution example to make your workflow easier to understand.

If your workflow uses one or more **skills**, your report should explicitly document them. In particular, you should provide the link to each skill, briefly explain what the skill is intended to do, and describe whether it is mainly instruction-only or whether it also relies on additional scripts, dependencies, or other external components. You should also comment on whether the agent was able to invoke the skill correctly in practice. If you modified the skill, adapted part of it, or changed the workflow in order to make it better fit your own task, you should describe those changes clearly.

The comparison should be a distinct part of the report. It should explain how the OpenClaw-based workflow differed from your earlier self-built LLM-agent workflow and what you learned from making that comparison. You should comment on both the workflow process and the final output, rather than discussing only one of the two.

## 5 Submission

Submit a reproducible record of your OpenClaw workflow. Depending on your implementation, this may include screenshots, exported configurations, supporting scripts if needed, or another runnable or inspectable record of how the workflow was set up. You should also submit a PDF report following the requirements described above.

A small but coherent and well-explained reimplementaion is preferred over a complicated but unstable one. You are not expected to build a large-scale production system. What matters most is that your submission demonstrates a clear use of **OpenClaw** to reimplement a previous NLP task and to compare the resulting workflow with an earlier self-built LLM-agent version.

### **Reminder to students:**

**If you did not complete Assignment 5, please ignored all requirements in this assignment that involve comparing your work with Assignment 5.**

**In addition, please note that the task you select must not duplicate the example demo presented in Lab 7.**